ZYQL: A Trivalent-Decision Quantum Language with Pre-Registered, Signed Evidence Receipts - Design, Formal Grammar, and End-to-End Validation on Superconducting Hardware

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ABSTRACT

We present ZYQL, a small quantum programming language whose distinguishing feature is an evidence discipline: every execution passes a purpose gate, produces a trivalent verdict (1, 0, or REFUSED), and is recorded in a hash-chained receipt that is pre-registered before data collection. We specify the language through a formal ISO/IEC 14977 EBNF grammar and small-step operational semantics; notably, the no-cloning theorem (T1) is enforced syntactically: a qubit binding may only be created from Dirac literals, never from another variable. We report a machine-validated grammar test battery (15/15), a compilation correctness proof by state fidelity (F = 1.000000000000 between the compiled circuit and the analytic target state), and two experimental campaigns on IBM Quantum superconducting processors: (i) a three-mission causal discrimination of error sources in Bell-correlation decay, culminating in a duration-matched experiment that identifies stochastic per-pulse error as the dominant mechanism (z = +3.61 against the duration hypothesis at matched circuit duration of 540,dt); and (ii) the first end-to-end execution of a complete ZYQL program (variational energy estimation of H_2) compiled from source and run on physical hardware (IBM ibm_fez, Heron r2), with tripartite verification across exact statevector, calibrated noise simulation, and hardware. All claims in this paper are labeled by a four-state maturity scale (IMPLEMENTADO / EXPERIMENTAL / ESPECIFICACAO / BLOQUEADO), and all deviations from preregistered acceptance criteria are reported, including a hardware VQE energy of -0.9295 pm 0.0105 Ha versus an exact FCI value of -1.136189 Ha (deviation 0.2066 Ha, unmitigated, versus a preregistered acceptance threshold of 0.20 Ha). Raw counts, preregistration certificates, and job identifiers are published for full reproducibility.

SUMMARY OF EVIDENCE

- Language: formal EBNF (ISO/IEC 14977) + small-step operational semantics; parser battery 15/15 machine-validated.
- T1 no-cloning enforced syntactically (a qubit can only be born from Dirac literals).
- Compilation proof: state fidelity 1.000000000000 (kernel IR -> OpenQASM 3 -> qiskit) vs analytic ansatz state.
- FRONTIER 001-003 (ibm_fez): causal discrimination of ZZ decay; duration-matched delay experiment decides stochastic per-pulse error (z=+3.61), job db5qqb4lf4us73c5gglg.
- QLAB-002 (ibm_fez): first complete .zeph program (VQE of H2, STO-3G, R=0.735 A, JW) compiled from source and run on hardware; tripartite verification; jobs db6k5e4lf4us73c5gglg, db6kassvf2bc73d1ca0g; 61,440 shots.
- Hardware VQE minimum: -0.9295 +/- 0.0105 Ha vs exact FCI -1.1361891625 Ha; deviation 0.2066 Ha UNMITIGATED - preregistered 0.20 Ha criterion FAILED by 0.0066 Ha and reported as such.
- All claims labeled: IMPLEMENTADO / EXPERIMENTAL / ESPECIFICACAO / BLOQUEADO. Raw counts, preregistered certificates and scripts published at github.com/SerenaSolutions/zephirum.